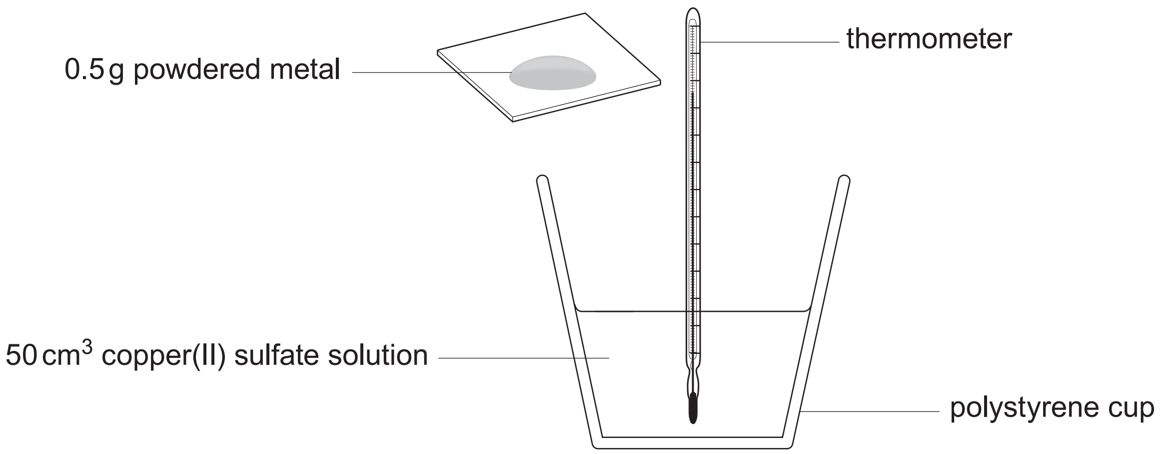
5. Joe, Alex and Megan were asked to investigate the temperature rise when four metals were added to excess copper(II) sulfate solution. 0.5 g of each metal was added to separate 50 cm^3 samples of copper(II) sulfate solution.

The temperature rise for each reaction was measured.

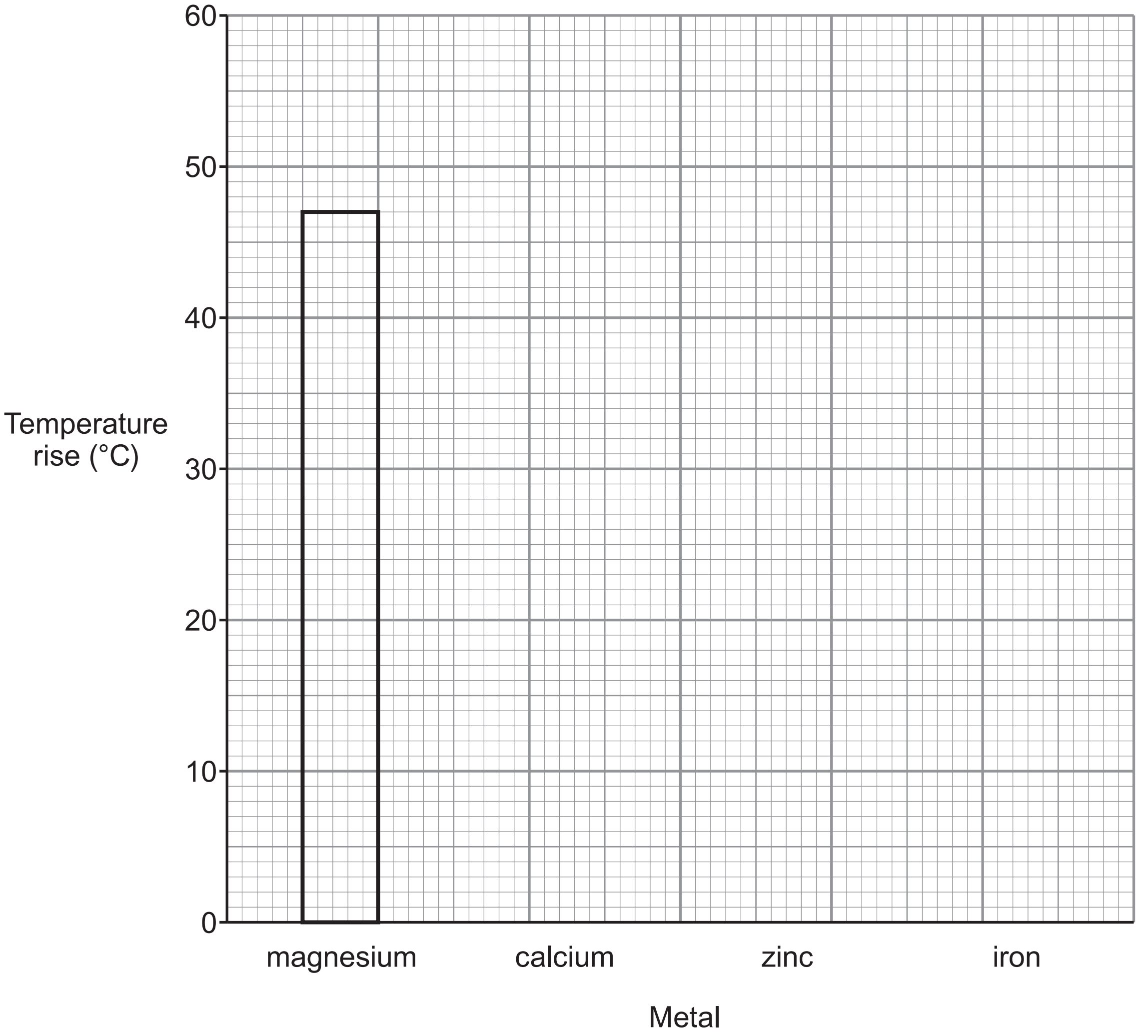


Their results are shown in the table below.

Metal	Temperature rise (°C)
magnesium	47
calcium	54
zinc	38
iron	25

Examiner only

(a) (i) Draw a bar chart on the grid to show the results. One bar has been drawn for you. [2]



(ii) Put the **four** metals in order of reactivity. [1]

Most reactive ....

....

....

Least reactive ....

Examiner only

(iii) The temperature rise for each metal was lower than expected. The students were asked to suggest how the apparatus could be improved so that the temperature rise recorded for each metal was closer to the expected value.

Joe suggested wrapping the polystyrene cup in cotton wool.

Megan suggested putting a lid on the polystyrene cup.

Alex suggested using a cup made from copper.

Choose which student's suggestion would **not** result in the temperature rises being closer to the expected value. Explain your choice. [2]

Student ....

Reason ....

....

....

(iv) Complete the equation for the reaction between magnesium and copper(II) sulfate solution. [2]

$Mg + CuSO_4 \longrightarrow$ .... + ...

Examiner only

(b) The students repeated the experiment using chromium and recorded a temperature rise of 30 °C.

What does the temperature rise of 30 °C tell you about the reactivity of chromium compared to the four metals in part (a)? [1]

....

....

(c) The energy given out by the reaction can be calculated using the formula below.

energy given out (J) = volume of solution (cm^3) × 4.2 × temperature rise (°C)

Calculate the energy given out during the displacement reaction between **iron** and copper(II) sulfate solution. [2]

Energy given out = ...J

10